Testing Pack

1. Test Plan

Authentication & Authorization Tests

Test ID	Test Case	Expected Result	Test Data
AUTH-01	Login with wrong password	Error: "Invalid credentials. Attempt X of 3"	admin1/wrongpass
AUTH-02	Login with correct admin credentials	Admin dashboard loads	admin1/admin123
AUTH-03	Login with instructor credentials	Instructor dashboard loads	sambuddho/sam123
AUTH-04	Login with student credentials	Student dashboard loads	stu1/student123

AUTH-05	3 failed login attempts	Account locked for 2 minutes with countdown	Any user/wrong password
AUTH-06	Verify password hash in Auth DB	No plain text passwords visible	Check `users` table
AUTH-07	Verify ERP DB has no passwords	Only Auth DB contains password_hash	Check all ERP tables

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2. Student Functionality Tests

Test ID	**Test Case**	**Expected Result**	**Test Data**
STU-01	View course catalog	Shows courses with code, title, credits, instructor, capacity	Login as stu1
STU-02	Register for available section	Success message + appears in "My Courses"	CS101-A (capacity available)

STU-03	Try to register same section twice	Error: "Already enrolled in this section"	CS101-A (already enrolled)
STU-04	Try to register full section	Error: "Section is full"	Section with 0 available seats
STU-05	Drop course before deadline	Success + removed from "My Courses"	Any enrolled section
STU-06	Try to drop after deadline	Error: "Drop deadline has passed"	Set deadline in past
STU-07	View grades for enrolled courses	Shows finalized grades only	Sections with FINALIZED status
STU-08	Download grades as CSV	CSV file downloads with correct data	Click "Download Grades"

STU-09	Try to view another student's data	Not accessible/blocked	Manual URL manipulation
STU-10	View timetable	Shows enrolled sections with schedule	My Courses page

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3. Instructor Functionality Tests

Test ID	**Test Case**	**Expected Result**	**Test Data**
INST-01	View only own sections	Shows only assigned sections	Login as sambuddho
INST-02	Add/Update student grades	Grades saved successfully	CS201-cv90, add quiz scores
INST-03	View student marks chart	Bar chart shows student performance	Manage Grades section

INST-04	Import grades from CSV	Grades imported + notifications sent	Upload valid CSV
INST-05	Download section grades as CSV	CSV downloads with all student data	Click "Download as CSV"
INST-06	Finalize grades (weight = 100%)	Success + students see final grades	Section with total weight 100%
INST-07	Try to finalize grades (weight ≠ 100%)	Error: "Total weight must be 100%"	Section with weight 80%
INST-08	Try to edit another instructor's section	Blocked with error message	Try to access other section
INST-09	View grades with non-finalized students	Only non-finalized students shown	After finalizing some students

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4. Admin Functionality Tests

Test ID	Test Case	Expected Result	Test Data
ADMIN -01	Create new student user	User created in Auth + ERP DB	Username: newstudent1
ADMIN -02	Create new instructor	Instructor created with profile	Username: newinst1
ADMIN -03	Create new course	Course appears in catalog	CS301, 4 credits
ADMIN -04	Create new section	Section created + instructor assigned	CS301-A, sambuddho
ADMIN -05	Delete section (no enrollments)	Section deleted successfully	Empty section

ADMIN -06	Try to delete section (has enrollments)	Error: "Students are enrolled"	Section with enrollments
ADMIN -07	Toggle Maintenance Mode ON	Banner shows + writes blocked	Settings page
ADMIN -08	Student tries to register (Maintenance ON)	Error: "System under maintenance"	Try course registration
ADMIN -09	Instructor tries to add grades (Maintenance ON)	Error: "System under maintenance"	Try grade entry
ADMIN -10	Toggle Maintenance Mode OFF	Normal operations restored	Settings page
ADMIN -11	View system statistics	Shows user counts, enrollments	Dashboard

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5. Data Integrity Tests

Test ID	Test Case	Expected Result	Test Data
DATA-01	Prevent duplicate enrollments	Error: "Already enrolled"	Try to enroll twice
DATA-02	Negative course capacity	Validation error	Try capacity = -5
DATA-03	Invalid credit values	Validation error	Try credits = 0 or negative
DATA-04	Grade weight validation	Total must equal 100%	Add assessments
DATA-05	Enrollment capacity check	Cannot exceed section capacity	Full section
DATA-06	Cascade delete check	Document behavior or block	Delete course with sections

6. Maintenance Mode Tests

Test ID	Test Case	Expected Result	Test Data
MAINT-01	Enable maintenance mode	`settings_bach.maintenance_mode = 'true'`	Admin toggle
MAINT-02	Student register blocked	Error shown across all reg screens	Try any registration
MAINT-03	Student drop blocked	Error message displayed	Try to drop course
MAINT-04	Instructor grade entry blocked	Error message displayed	Try to add grades
MAINT-05	Admin can still make changes	Admin operations work	Create user/course

MAINT-06	Maintenance banner visible	Shows on all user dashboards	Login as any role
MAINT-07	Disable maintenance mode	All operations restored	Admin toggle OFF

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7. Security Tests

Test ID	Test Case	Expected Result	Test Data
SEC-01	Password stored as hash only	BCrypt hash in `users.password_hash`	Check Auth DB
SEC-02	SQL injection attempt	Query sanitized/blocked	Login: `' OR '1'='1`
SEC-03	Session hijacking prevention	Session tokens validated	Manual session manipulation

SEC-04	Cross-user data access	Blocked at service layer	Try to access other user data
SEC-05	Forgot password with OTP	OTP sent + password reset works	Use Firebase OTP flow

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8. UI/UX Tests

Test ID	Test Case	Expected Result	Test Data
UI-01	Error messages are user-friendly	Clear, non-technical messages	Various error scenarios
UI-02	Success messages displayed	Green/positive feedback shown	Successful operations
UI-03	Tables display properly	Data formatted correctly	Course catalog, grades

UI-04	Buttons have clear labels	Icons + text descriptive	All dashboards
UI-05	Responsive layout	UI adapts to window resize	Resize window
UI-06	Loading indicators	"Please wait" shown for operations	Long operations
UI-07	Background image/transparency	Login screen displays correctly	Login page
UI-08	Marquee text animation	Welcome text scrolls smoothly	Login page

9. Performance Tests

Test ID	Test Case	Expected Result	Test Data
PERF-01	Load 100+ courses in catalog	Loads within 3 seconds	Seed 100 courses

PERF-02	Application startup time	Opens within 5 seconds	Fresh start
PERF-03	Grade chart rendering	Chart loads within 2 seconds	30+ students
PERF-04	CSV export generation	File ready within 5 seconds	Large dataset